FATAYO MICHAEL AKOREDE

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RESEARCH INTERESTS

Real-time machine learning systems; fraud and anomaly detection; imbalanced classification; causal feature engineering and leakage-free evaluation protocols; streaming data architectures; trustworthy and explainable machine learning for financial systems.

EDUCATION

Master of Science (MSc), Computer Science Lead City University, Ibadan, Nigeria
Final semester; thesis defense pending, expected 2026. Thesis: "Development of a Fault-Injected Reference Implementation of a Real-Time Fraud-Scoring Pipeline for Mobile Money Transaction Streams, with an Evaluation on PaySim."

Bachelor of Science (BSc), Computer Science Imo State University, Owerri, Nigeria
2020 – 2024. Second Class Honours (Upper Division); GPA 4.27/5.00.

RESEARCH EXPERIENCE

FraudShield: Real-Time Fraud Detection Research Lead City University
MSc Thesis Research; independent researcher | Code: github.com/mafatayo/fraudshield (release v1.0.2, doi.org/10.5281/zenodo.23297120)

- Designed a real-time architecture for fraud scoring in mobile money transaction streams (Kafka, Spark, Redis and PostgreSQL specified at design level) and built a reference implementation of the scoring pipeline with local stores. It processed all 6.36 million PaySim events at about 4,700 events per second with a 99th percentile latency of 0.79 ms, and kept its guarantees under 1,304 injected process crashes.
- Evaluated XGBoost, a random forest, logistic regression and hand-written rules under random and temporal splits, with five seeds and bootstrap intervals. A 23-feature XGBoost model catches 99.96% of frauds at zero false alarms, but its key inputs are balances recorded after posting, so it is a post-posting monitor; a 19-feature pre-authorisation configuration reaches PR-AUC 0.9997 (XGBoost) and 0.9999 (random forest).
- Showed that the benchmark is too easy to rank models: a hand-written rule reaches 97.5% recall with no false alarm, a depth-3 tree 99.7%, and the two ensembles cannot be separated on ranking quality. Reported the headline figures as an upper bound, and found that velocity features add nothing on the full set, so the state protocol guards a feature family PaySim cannot reward.

Software Quality Assurance Process: A Consolidated Definition and a Lifecycle-Mapped Taxonomy of Tool Support Lead City University
Independent research paper, SEN 712 (Software Quality Assurance)

- Conducted a structured narrative review across IEEE Std 730/1028/829, ISO/IEC 25010, and the SWEBOK Guide to resolve persistent terminological conflation between quality assurance, quality control, and testing, producing a consolidated working definition of the SQA process.
- Constructed a lifecycle-mapped taxonomy of QA tool categories (requirements traceability, static analysis, review support, testing frameworks, CI, defect tracking) to replace vendor-driven tool selection with a lifecycle-stage-driven rationale; proposed empirical validation of the taxonomy against the FraudShield pipeline as future work.

PUBLICATIONS & PRESENTATIONS

Fatayo, M. A. "Auditing Feature Reliance for Adversarial Robustness in Real-Time Fraud Detection." Submitted, ICAIF 2026 workshop on Financial AI Security, Privacy, and Safety, Milan, Nov. 2026 (decision pending).

Fatayo, M. A. “FraudShield: A Real-Time Machine Learning Framework for Fraud Detection in High-Volume Financial Transaction Streams.” Submitted, Bridges (5th International Conference on Frontiers in Multidisciplinary Science and Artificial Intelligence, FASCON 2026), Lead City University, Ibadan, Nov. 2026.

Conference abstract submitted, FASCON 2026 (Faculty of Natural and Applied Sciences, Lead City University), Ibadan, Nigeria, Nov. 2026.

Fatayo, M. A. “Software Quality Assurance Process: A Consolidated Definition and a Lifecycle-Mapped Taxonomy of Tool Support.” Preprint, Zenodo, Sept. 2026. https://doi.org/10.5281/zenodo.22341782

Fatayo, M. A. “Replication package and preregistration: lifecycle coverage of QA tooling and user-reported bugs in open source projects.” Zenodo, Oct. 2026. https://doi.org/10.5281/zenodo.23184163

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C, SQL.

Machine Learning & Data: pandas, scikit-learn, XGBoost, imbalanced-classification methods, causal feature engineering, model evaluation (PR-AUC, ROC-AUC).

Systems: streaming pipeline design (Kafka, Spark Streaming, Redis, PostgreSQL at design level), fault injection, and latency benchmarking.

Full-Stack Development: React, Node.js, MongoDB, Socket.io.

PROFESSIONAL EXPERIENCE

Founder & Lead Developer, Khor
Built and deployed a full-stack SaaS platform for gym management (React, Node.js, MongoDB, Socket.io), including a real-time QR check-in system, Paystack split payments between gyms and the platform, and revenue-tracking dashboards.

Chief Technology Officer, Khor Consolidated Ile-Ife, Osun State | Sept 2020 – Dec 2023
Managed technical infrastructure and content platform operations.

Front End Developer, Upwork Global Inc. Remote | Apr 2021 – Sept 2025
Developed responsive, browser-compatible web applications and e-commerce platforms for international clients.

CERTIFICATIONS & HONORS

Software Engineering Certification, ALX Africa (2023).

AI Programming with Python Nanodegree, Udacity (2024).

Second Class Honours (Upper Division), BSc Computer Science, Imo State University (GPA 4.27/5.00).